



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP2523 – DATABASE

SEM 1 - 2024/2025

INITIAL SYSTEM PROPOSAL

**KADA REGISTRATION
SYSTEM**

Group Name: Technicrab

Members:

- | | |
|------------------------------------|-----------|
| 1. Cheryl Cheong Kah Voon (Leader) | A23CS0060 |
| 2. Chau Ying Jia | A23CS0213 |
| 3. Neo Li Xin | A23CS0253 |
| 4. Lau Yee Wen | A23CS0099 |
| 5. Woo Cheng Shuan | A23CS0283 |

Stakeholder: Koperasi Kakitangan Kada Kelantan Berhad

Representative Name	Date Interviewed
Mr. Ahmad Rohailin bin Hani	14/05/2024

PROPOSED SYSTEM BACKGROUND

1. Introduction

In an age where efficiency and security are paramount, organizations must adapt to meet the demands of modern users while ensuring operational excellence. For Koperasi Kakitangan KADA Kelantan Berhad, a cornerstone of financial support and cooperative services, this transformation begins with reimagining how its core processes are managed.

We are honored to introduce a KADA Registration System. The main goal of the proposed KADA Registration System is to create a fully integrated digital platform. This system can enhance the efficiency, security, and user experience for all members and administrative staff. The system seeks to replace the current manual processes, which are time-consuming, error-prone and highly human resource-dependent jobs by replacing present manual processes. The key features of our system include online membership registration, real-time application status tracking, automated financial statements, and secure digital storage of sensitive data.

By simplifying and improving their individual responsibilities within the cooperative, members, administrative staff, and Board Directors are the three main user groups that the KADA Registration System benefits. Members may track loan applications, payments, savings, and dividends with real-time updates and automated notifications through an easy-to-use, secure online registration and service portal. The workload and mistakes that come with manual processing are decreased when administrative staff have a centralized platform to monitor and check applications, approve loans, and keep an eye on digital records. The system provides Board Directors with an accountable and transparent means of reviewing important applications and reaching strategic choices that support KADA's objectives. When combined, these enhancements produce a productive, safe, and easy-to-use environment that boosts member happiness and productivity throughout the company.

Ultimately, the goal of the KADA Registration System is to create a safe, effective and user-friendly platform that strengthens security with features, improves data management and facilitates communication. By automating key processes and providing a seamless experience for all users, the system aims to improve service delivery, boost productivity, and foster member satisfaction, driving KADA's long-term success.

The objective of the KADA Registration System project is to increase productivity, lower human error rates, and improve user experience. The system aims to digitize and streamline the cooperative's member registration, loan application, and financial management procedures. The technologies applied seek to decrease processing times, give members real-time information, and facilitate improved decision-making for board directors and administrative staff by putting automated workflows and safe data handling into place. In the end, this project aims to provide a more effective, accessible, and secure digital platform in order to promote increased productivity and member satisfaction.

2. The Existing System

The existing system operates manually, which limits its efficiency and productivity. Nevertheless, it does offer several main services and features, as listed below:

1. Membership Registration

This is a structured process for KADA's staff to become part of their members, granting them access to a variety of cooperative benefits and services. The current member registration system relies on a manual process, necessitating individuals to complete information on printed forms. During registration, various personal details, such as name, identification number, address, and others are gathered. Afterwards, KADA staff will need to review the registration documents again. This method is not only time-consuming but also susceptible to errors and places a significant strain on human resources.

KOPERASI KAITANGAN KADA KELANTAN BERHAD
PERMOHONAN MENJADI ANGGOTA
MAKELUHATI PENCAMUK

NAMA: _____ JANTINA ☐ LELAKI ☐

NO. KP: _____ PEREMPUAN ☐

TARAF PERKAHWINAN: _____ AGAMA: ☐ ISLAM ☐ LAIN-LAIN ☐

ALAMAT RUMAH: _____ Negeri: _____

POSKOD: _____ BANGSA: ☐ MELAYU ☐ LAIN-LAIN ☐

NEGERI: _____ Negeri: _____

NO. ANGGOTA: _____

NO. PI: _____

JAWATAN & GRED: _____

ALAMAT PEJABAT: _____

(Tempat Bertugas): _____

POSKOD: _____

NO. TEL/FAX: _____

NO. TEL BIMBIT: _____

NO. TEL RUMAH: _____

Gaji bulanan: RM _____

BAHAGIAN KELOMPOK DAN PERAWAK

BIL	HUBUNGAN	NAMA	NO. K/P @ NO. SRT BERKAWAT

* SILA BUAT LAPORAN JIKA BUKAN YAKSI DISEDIAKAN TODAY MENCURUP

YURAN DAN SUMBANGAN

JIKA DI TERIMA SEBAGAI ANGGOTA, SAYA BERSITUJU MEMBAYAR YURAN DAN SUMBANGAN BULANAN SEPERTI DI BAWAH :

BIL	PERKARA	RSE
1	FEE MASUK	
2	MODAL SYER *	
3	MODAL YURAN	
4	WANG DEPOSIT ANGGOTA	
5	SUMBANGAN TABUNG KESALAHAN (AL-ASRAR)	
6	DIMPANAN TETAP	
7	LAIN-LAIN	

* SUMBANG MODAL SYER AGALAH SEBANYAK RM 200.00 DARI TIDAK MELAKUKAN S.T. DARI MODAL SYER KOPERASI DAN MENGAJUKAN DUA LAKSANA DALAM TEMPAH 6 BULAN DARI TARIKH MELAKUKAN MENJADI ANGGOTA

Figure 3.1 Printed Member Registration Form

2. Loan Applications

Their loan application seeks financial assistance for personal, educational, or business needs. The goal of the loan application process is to be clear and efficient, allowing cooperative members to obtain the required funds quickly. Currently, the loan application system is manual too, necessitating that information be entered on printed forms. In addition, the qualified members are not able to check on their past transactions and records it is hard for them to look back for their total loans and payments if they lost their statements.



KORPRI
KOPERASI KARTUNGAN KADA KELANTAN BERHAD



KORPRI
KOPERASI KARTUNGAN KADA KELANTAN BERHAD

Borang KKKR 02/00

PERMOHONAN PEMBIAYAAN ANGGOTA

RUTIR-BUTIR PEMBIAYAAN

JENIS PEMBIAYAAN

☐ AL-SAM	☐ BAKI PUJUH PENGIRAAN
☐ AL-IMDAH	☐ CUKAI KALIAN
☐ SEMU KUALA	☐ LAIN-LAIN
☐ KARNIVAL MUSEM STRIMOH	Nota lain _____

AMALAH DIPONON

RM +

TEMPON PEMBIAYAAN

ANGKARAN BULAN _____

BULAN

RM +

RUTIR-BUTIR PERIBADI PEMOHON

NAMA

NO. KP

TARIKH LAHIR

ALAMAT RUMAH

POSKOD

NO. ANGKUTA

NO. HP

KAJANGAN

ALAMAT PEJABAT

(Tempat Berhajat)

POSKOD

NO. TEL. RUMAH

NO. TEL. BISNES

NAMA BAPA/IBU

NO. KAJUAN BAPA

JANTINA

☐ ISLAM
☐ NON-ISLAM

ASAMA

☐ ISLAM
☐ LAIN-LAIN

BERANDA

☐ ISLAM
☐ LAIN-LAIN

Nota lain _____

PENGALAMAN PEMOHON

Saya No. K.P.

ingin ini memohon kuasa

kepada KOPERASI KARTUNGAN KADA KELANTAN BHD atau mana-mana yang sah untuk mendapatkan apa-apa maklumat yang diperlukan

dan juga mendapatkan bayaran balik dari pengumpul dan dan membolehkan saya menggunakan semua yang diperuntukkan. Saya juga

berfikir untuk menerima pinjaman tambahan dari KOPERASI ini untuk membiayai perniagaan saya/ membiayai pendidikan/ atau lain-lain.

Tandatangan _____

Tarikh : _____

[illegible]

Figure 3.2 Member Loan Applications Form

3. Member Financial Statements

Koperasi Kakitangan KADA Kelantan Berhad currently uses a manual system to generate and distribute member financial statements. This involves collecting financial data manually from various records and compiling it into individual member statements. These statements include details such as the member's name, IC number, shares held, loans taken, and savings. Once the statements are prepared, they are reviewed by staff to ensure accuracy. After verification, the statements are handed out to members. Members then sign and return the verification portion to confirm receipt and accuracy. This manual process is time-consuming, labor-intensive, and prone to errors due to manual data entry. It also places a significant burden on the cooperative's limited human resources and delays members' access to their financial information. Implementing an automated system would streamline this process, reduce staff workload, and provide members with timely and accurate financial information.

Koperasi Kakitangan KADA Kelantan Berhad

NAMA: MOHD ROSLAN B ISMAIL
NO. K/P: 740807-03-5479 **NO. PF: 2273** **NO. AHLI: 1579**

Tuan/Puan,

PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023.

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah sebagaimana berikut:

MAKLUMAT SAHAM AHLI:

Modah Syer:	Modal Yuran:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM 0.00	RM 0.00	RM 0.00	RM 0.00	RM 0.00

MAKLUMAT PINJAMAN AHLI:

Al-Bai:	Al-Innah:	Bi-Pulih Kenderaan:	Road Tax & Insuran:	Khas:	Al-Qadriul Hassan:
RM0.00	RM0.00	RM0.00	RM0.00	RM0.00	RM0.00

PENGESAHAN BAGI PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023

Saya **MOHD ROSLAN B ISMAIL** No. Ahli: **1579** mengesahkan bahawa
Penyata Kewangan Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir 30 Jun 2023 adalah benar:

Tandatangan: _____
Tarikh: _____

☐ Setuju
☐ Tidak Setuju (Nyatakan sebabnya)

Nota:
1. Sila tandakan (✓) di salah satu kotak diatas.
2. Sekiranya pihak Koperasi tidak menerima sebarang maklumbalas daripada tuan/puan sehingga 30 Jun 2023, maka pengesahan penyata ini dianggap betul dan tuan/puan bersetuju.

Figure 3.3 Member Financial Statement

Problem with Existing System

The registration system at KADA company is currently handled manually, showing signs of low efficiency and productivity, as described below. Even though it offers only a limited number of essential services and features, there is room for improvement. Recognizing the constraints of the existing system emphasizes the necessity for a complete overhaul to enhance the system. Moving towards a new computerized system could greatly elevate their services and features. This new system will bring about automation, streamlined processes, an improved website for better performance, and an expanded array of services for users.

1. Insufficient registration process

In the current system, all registration procedures are manually processed, like customers filling out paper forms to register and staff handling each form individually. Personal information, both private and sensitive, is predominantly stored in hard copies during registration. This method not only poses security risks but also makes it challenging to retrieve or locate information when necessary. Additionally, the registration papers require further collection, rearrangement and scrutiny by staff at the end. It really takes much more time than doing the registration online.

2. Inconvenient Status Checking

In the existing process, members are unable to check the status directly, they must either call or wait for an email from the company to obtain the current status, whether it is about the registration approval or payment status. This method is not that convenient, as staff have to manually send emails instead of having an automated system on their website that segregates the status. Improvements are necessary to modernize this outdated system to enhance the efficiency of the processes.

3. Heavy Dependence on Human Resources

The existing manual system faces an issue as it heavily depends on human resources. All the steps starting from the registration and application for the loan are done manually. If the number of human resources decreases beyond a certain point, it will impact productivity and efficiency in the workplace. To address the issue of heavy dependence on human resources, the implementation of automation solutions is crucial as it can help streamline processes, minimize errors, and free up employees to focus on more strategic tasks that require human intervention.

4. Inefficient and human error

We know that retrieving and managing physical documents is time-consuming and the manual system heavily depends on human resources. This might lead to the chances of human error rising when staff forget to email the current status, missing payment receipts, or failing to gather customer information and also missing information. These inefficiencies will cause delays in service delivery and mistakes in data processing, affecting overall customer satisfaction and also might disrupt the entire development process as incomplete or inaccurate data may result. Therefore, the company requires a more well-designed system.

5. Security Concerns of Manually Stored Customer Forms and Information

In the digital age, safeguarding customer information is paramount for maintaining trust and ensuring compliance with data protection regulations. The cooperative currently

relies on manual storage of customer forms and information in physical cupboards. This method poses significant security risks that need urgent attention to protect sensitive data and uphold our commitment to customer privacy. As we can see, employees, cleaning staff, and visitors may easily access the cupboards, potentially leading to data breaches. Furthermore, it is challenging to track who accessed or modified the physical documents, making it difficult to audit and ensure the accountability of data handling practices.

Features	Existing System	Proposed System
Membership Registration	Yes	Yes
Loan application	Yes	Yes
Application status checking	No	Yes
Check financial status and statements	Yes	Yes
Check monthly and annually report	No	Yes
Check profit and loss from account	No	Yes

3. The Proposed System

The proposed KADA Registration System is a fully integrated digital platform aimed at enhancing the efficiency, security, and overall user experience of Koperasi Kakitangan KADA Kelantan Berhad. It replaces the current manual processes, which are time-consuming, prone to errors, and heavily reliant on human resources.

The system allows new users to register online by submitting personal details such as name, address, and identification data. Upon submission, this data is processed automatically, verified for accuracy, stored securely in a digital database, and provides real-time feedback on the status of the application, whether it is successful, pending, or rejected. Once a user becomes a registered member, they gain access to several convenient services, such as checking the real-time status of their registration, loan applications, and payments through an online portal. Members can also manage their savings, view financial statements, and track potential dividends from their investments, receiving automated email or SMS notifications regarding their financial activities.

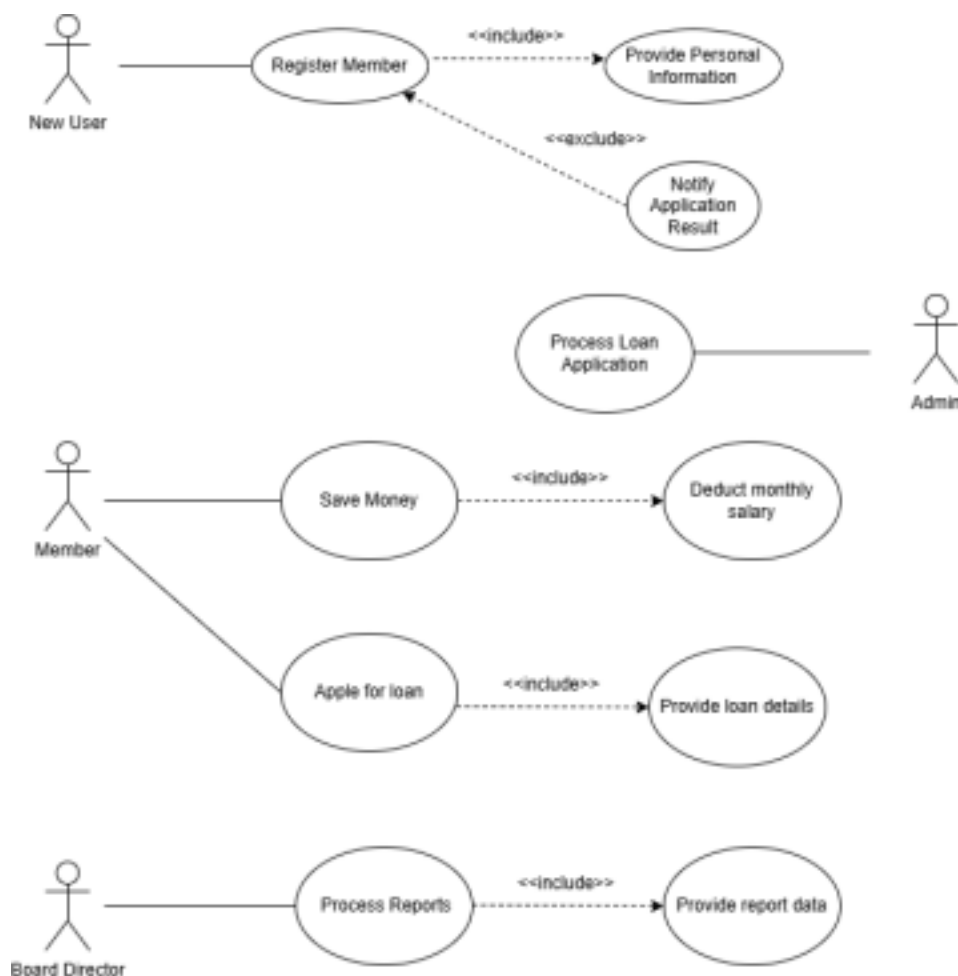
For the administrative staff (KADA employees), the system reduces the burden of manual data handling and processing. Admin is responsible for reviewing member applications, verifying submitted information, and authorizing loan applications based on pre-set criteria. For the admin, the system simplifies tasks by providing a centralized platform for reviewing member applications and loan requests. Admins can verify application details, approve or reject submissions based on eligibility criteria, and manage digital records securely, reducing the need for paper-based processes. They can also update the status of applications and ensure that members are promptly notified of any changes. The system minimizes human error by automating many of the routine processes, freeing up staff to focus on more strategic and high-priority tasks.

Board Directors, who oversee high-level decisions, are involved in reviewing and approving larger or more critical loan applications. Their role is to ensure that these applications align with the cooperative's strategic goals and regulatory requirements. They have the authority to make final decisions on both member and loan applications, with their approval or rejection logged into the system for transparency and future reference.

To develop the KADA Registration system, several technologies should be implemented to optimize the system's performance, security and scalability. Several technologies should be implemented in helping in the development of this system. First of all is the HTML language in structuring and coding the content of the website interface. Next, the JavaScript using for designing the backend system. Bootstrap for front-end framework especially for mobile interface. Lastly, the MySQL database for structured data storage.

Overall, the KADA Registration System brings about a more time-saving, secure, and user-friendly environment for all stakeholders. It reduces the burden on human resources, improves data management, enhances communication, and ensures that sensitive member information is stored securely. Also, it enhances security measures through features like Two-Factor Authentication (2FA) and provides a seamless experience for users. By automating key functions and providing a streamlined user interface, the system aims to elevate the cooperative's service delivery and operational capabilities, fostering greater member satisfaction and organizational productivity.

Use Case Diagram for Proposed System



4. Glossary

- **KADA**

An agency under the Ministry of Agriculture and Food Security that is responsible for promoting and encouraging the economic and social development of farmers in the Kemubu area. It strives to improve the socio-economic status of farmers and the country's food production.

- **HTML**

HyperText Markup Language will be used for structuring and coding the content of the KADA Registration System's web interface. Our team is using HTML to design the interface of the website. This programming language provides the foundation for creating and organizing the layout and elements of the user interface, ensuring clear and accessible content display for users.

- JavaScript

JavaScript is used at the backend for this system to handle server-side logic and processing. It is an essential part of any modern website that can make a huge difference on the user experience and functionality. JavaScript enables responsive and efficient server operations, processing member registrations, loan applications, and data requests smoothly.

- Bootstrap

Bootstrap is a front-end framework that provides responsive, mobile-first design capabilities. It will be used to create a user-friendly and consistent mobile interface for the KADA system, enabling members to access their accounts and information seamlessly on various devices, including smartphones and tablets.

- MySQL

Relational database management system for structured data storage. It will handle all member records(eg. personal details), their financial information (eg. statements and reports), and loan application data (eg. application status) details. MySQL guarantees data integrity, facilitates efficient querying and supports complex data relationships, making it an ideal choice for managing the data needs in the KADA system.

5. Bibliography

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